

Another is that pessimists often extrapolate present trends without accounting for how markets adapt.

In 2008 environmentalist Lester Brown wrote: “By 2030 China would need 98 million barrels of oil a day. The world is currently producing 85 million barrels a day and may never produce much more than that. There go the world’s oil reserves.”⁵⁹

He’s right. The world would run out of oil in that scenario.

But that’s not how markets work.

There is an iron law in economics: extremely good and extremely bad circumstances rarely stay that way for long because supply and demand adapt in hard-to-predict ways.

Consider what happened to oil immediately after Brown’s prediction.

Oil prices surged in 2008 as growing global demand—much of it from China—crept up to potential output. A barrel of oil sold for \$20 in 2001 and \$138 by 2008.⁶⁰

The new price meant drilling oil was like pulling gold out of the ground. The incentives for oil producers changed dramatically. Hard-to-tap oil supplies that weren’t worth the fight at \$20 a barrel—the cost of drilling didn’t offset the price you could sell it for—became the bonanza of a lifetime now that you could sell a barrel for \$138.

That sparked a surge of new fracking and horizontal drilling technologies.

The Earth has had roughly the same amount of oil reserves for all of human history. And we’ve known where the big oil deposits are for some time. What changes is the technology we have that lets us economically pull the stuff out of the ground. Oil historian Daniel Yergin writes: “86% of oil reserves in the United States are the result not of what is estimated at time of discovery but of the revisions” that come when our technology improves.

That’s what happened as fracking took off in 2008. In the United States alone oil production went from roughly five million barrels per day in 2008 to 13 million by 2019.⁶¹ World oil production is now over 100 million barrels per day—some 20% above what Brown assumed was the high mark.